EART11300 practical: Igneous, sedimentary and metamorphic rocks in hand specimen

By the end of the practical you should be able to:

- Recognise that a particular rock is igneous, sedimentary or metamorphic
- Further classify rocks using grain size
- Practise sketching rock samples
- In some cases, identify the minerals that a rock contains

The PDFs of the 3 videos on Blackboard contain useful background information for this practical. These videos explain how to recognise (1) igneous rocks, (2) sedimentary rocks, and (3) metamorphic rocks.

There are 15 hand specimens for you to look at.

1. Using the hand specimen attributes listed below (A), and the suggested steps to follow (B), assign each sample to one of the three rock types: igneous, sedimentary or metamorphic. Fill in the table on page 3 showing the criteria you used to classify them.
2. When you have determined the rock type, try to classify them further according to the criteria on pages 4-5 (C).
3. Draw a sketch of one of the following samples: R14 or S77. You do not need to draw the whole sample, but it needs to be enough to show what kind of rock it is. For example, if it is a sedimentary rock that contains clasts, draw a few clasts and the surrounding matrix, and make sure that, as best as possible, you show the range of shapes and sizes of the clasts and the correct clast:matrix proportions. Your sketch should include a title, a scale bar and some annotations.
4. In some of the samples you may be able to identify one or more minerals from last week: quartz, calcite, feldspar, mica (biotite, muscovite). Where crystals are large enough to see them clearly, try to identify them.

A. Attributes of the three categories of rock in hand specimen or thin section:

Igneous Rocks

These rocks have crystallised from partially molten material – magma – either within the Earth or on Earth's surface.

When you are looking at a hand specimen or thin section:

- Igneous rocks tend to have interlocking crystals
- Some crystals have their own shape (i.e. are bounded by flat crystal faces) indicating unhindered growth from a liquid.
- Contain relatively few minerals (normally 5 different mineral species or less).
- Crystals of each mineral tend to be quite uniform.
- May contain fossil gas bubbles (= vesicles). Not otherwise porous.
- Crystals don't usually have a preferred orientation (but there are exceptions, e.g. cumulates and rocks with flow-aligned crystals)
- Do not contain organic matter or fossils.

Sedimentary rocks

These rocks have formed by the accumulation, as layered deposits on the Earth's surface, of material derived by weathering of pre-existing rocks or material of biological origin.

When you are looking at a hand specimen or thin section:

- Commonly consist of particles (clasts) of pre-existing rocks, which may include polycrystalline rock fragments.
- Clasts may be rounded, indicating abrasion during sediment transport.
- May contain fossils.
- May contain organic matter.
- May be quite porous.
- Commonly layered (bedded), often with dramatic compositional contrasts. (May only see this when looking at a large scale, e.g. on maps or in the field.)

Metamorphic rocks

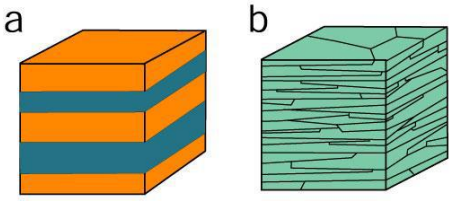
These rocks have formed by alteration of pre-existing igneous or sedimentary rocks, <u>in the solid state</u>, through the action of heat and/or pressure.

When you are looking at a hand specimen or thin section:

- Compositional layering (fossil bedding) may be present, but may be deformed (folded, etc).
- Contain relatively few minerals (normally 6 or less); crystals of each mineral tend to be quite uniform.
- Crystals often have a preferred orientation (tectonic fabric) produced by deformation. If the fabric is planar (foliation), it may be discordant to any layering present.
- Some crystals have their own shape; others have shapes indicative of competitive growth in the solid state (analogy: bubbles in foam).
- No significant porosity.
- No fossils usually survive.

Difference between (a) compositional layering and (b) foliation:

Layers have different compositions



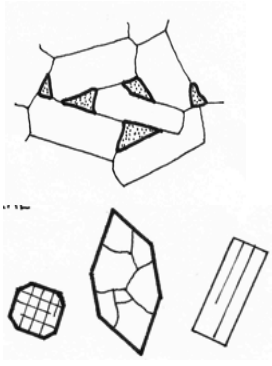
Flaky minerals are all in the same orientation

B. Steps to follow to help you classify the samples:

(a) Is it made of clasts (particles) of pre-existing rock (including fossils)? If **yes**, then sedimentary.

(b) If **no**, and you can see some or all of the crystals:

- Are the crystals interlocking, e.g. looking something like this…. with some of them having their own shape? If **yes**, then likely to be igneous. You will probably <u>not</u> see any layering or preferred orientation. Having their own shape looks like this…..



- Do **few** crystals have their own shape? If **yes**, then likely to be metamorphic. You are likely to see layering or preferred orientation (but it may be that you don't see either layering or preferred orientation).

(c) If **no** (or you can't tell), and most of the crystals are **too small** for you to see their shapes:

- If it contains fossils or organic matter, then it is sedimentary.
- If the minerals have a preferred orientation (foliation), then it is metamorphic. (The foliation will mean that the rock can split into layers, and you will probably see a sheen on the surface where all the platy minerals are lined up.)
- If it is not foliated, with no fossils or organic matter, but there are a few large crystals with their own shape, then it is igneous.
- If it looks glassy, then it is igneous.

Rock	Made of clasts? yes/no	Can you see interlocking crystals? y/n	Crystals have their own shape? y/n	Compos-itional layering? y/n	Foliation ? y/n	Fossils or organic matter? y/n	Glassy? y/n	Pores or vesicles? y/n	Ig/Met/Sed
L106									
R5									
R14									
R17									
R26									
R31									
R45 or A1									
R47									
S77									
S110									
W12									
X									
Z									
Φ1 or Φ2									
Φ4 or Φ5									

C. Further classification of the different rock types:

Igneous Rocks.

A simple classification of igneous rocks is based upon the grain size, which tells us how quickly the rock has cooled:

- Medium- to coarse-grained = slow cooling = **plutonic**
- Fine-grained = fast cooling = **volcanic**

Note that if the rock contains larger crystals in a finer-grained "groundmass" (= the rock matrix), then this indicates a 2-stage cooling process, with the larger crystals growing slowly in a magma chamger, and the groundmass crystallising in the eruption.

In the following table, indicate whether the igneous rocks are plutonic or volcanic.

Plutonic	Volcanic

Sedimentary Rocks.

If the rock is made of particles, then it can be classified according to grain size, which indicates the distance travelled from source to deposition:

- **Proximal** = close to source = coarser grain size.
- **Distal** = far away from source = finer grain size.

The grain size reflects the amount of energy available for transportation. This classification does not work so well for rocks containing fossils whose size is determined by the kind of fossil rather than distance travelled.

In the following table, arrange your sedimentary samples into 3 categories from closest to source (coarsest grain size) to furthest away from source.

	Sample numbers
Closest to source	
Furthest from source	

Metamorphic Rocks.

Can be classified according to the extent that the rock has changed during metamorphism, which usually depends on the temperature reached, and is known as **metamorphic grade.**

- **Low grade** display the least amount of change. If the rock started out as fine-grained (e.g. a mudstone), then low-grade metamorphism will result in a rock that is still fine-grained.
- **Medium grade** show an intermediate level of change. It will be coarser grained.
- **High grade** show the most change. It will have the largest crystals.

In the following table, arrange your metamorphic samples into 3 categories from lowest grade (finest grain size) to highest grade. Note that it can be difficult to compare the metamorphic grade of rocks with very different compositions.

	Sample numbers
Lowest grade	
Highest grade	